

Problem Statements

TRACK 1 · FORENSIC SCIENCE

Physical & Digital Evidence

01 AI-Powered Document Forgery Detection Assistant

Real Case: Fake COVID vaccination certificates sold across India in 2021 (Telangana, UP). Education Ministry recorded 3,000+ fake degree cases in 2022. Courts regularly receive forged property documents and altered FIRs.

Build a Bob-powered guided examination workflow where a forensic document examiner inputs structured observations about a suspected forged document — font inconsistencies, signature mismatches, paper anomalies, ink spread, digital metadata flags. Bob classifies the anomaly type, assigns a forgery confidence level, maps findings to examination standards, and drafts a court-admissible expert opinion report.

02 Disaster Victim Identification (DVI) Coordinator

Real Case: Odisha Balasore train collision (June 2023, 293 deaths) — NDRF teams manually cross-referenced 100+ unidentified bodies against missing person reports over 72 hours. No unified matching system existed on-ground.

Build a Bob-powered DVI coordination tool where teams log ante-mortem profiles (family-provided: physical description, scars, birthmarks, clothing, dental notes) and post-mortem observations. Bob cross-references all entries, produces a match-probability score for each pairing, surfaces the top 3 candidates per unidentified body with rationale, and generates a reconciliation report ready for forensic confirmation.

03 AI-Based Crime Scene Evidence Prioritization

Real Case: 2019 Hyderabad veterinarian murder — investigators received 3,000+ photos and 200+ physical items. Without prioritization, a DNA-bearing cigarette butt was nearly missed in the first sweep. Overloaded FSLs delay testing by weeks.

Build a Bob-powered evidence triage assistant where an investigator describes or lists crime scene items. Bob classifies each by evidence category, assigns a forensic priority score based on crime type and probative value, flags items requiring urgent specialized testing, and outputs a ranked examination schedule for FSL submission.

04 Deepfake Detection for Criminal Investigations

Real Case: Deepfake video of actress Rashmika Mandanna went viral (Nov 2023). Mewat sextortion gangs used AI-generated blackmail videos targeting thousands. Cyber cells have no structured analysis framework or legal packaging tool for such cases.

Build a Bob-powered cyber cell investigation assistant that guides an officer through a structured deepfake analysis checklist (metadata inconsistencies, compression artifacts, facial landmark anomalies, audio-lip sync discrepancies). Bob documents findings, generates an evidence summary, maps the case to IT Act 2000 / BNS sections, and produces an internal investigation brief and a victim-facing reporting guide.

05 Cyber Fraud Network Analyzer

Real Case: Jamtara SIM-swap ring (Jharkhand) — 95,000+ UPI fraud cases in FY2023. Investigators manually traced connections between bank accounts, SIM cards, and devices over weeks. Most cases went unsolved due to lack of network visualization tools.

Build a Bob-powered investigation tool that takes unstructured cyber fraud intelligence (transaction records, call logs, device IDs, accused names — mock inputs), extracts entities and relationships, identifies the fraud pattern type, maps the organizational hierarchy (kingpin → mule → victims), and generates an FIR-ready case brief with recommended actions.

06 Social Media Threat Intelligence Engine

Real Case: The 2022 Nupur Sharma controversy and 2020 Delhi riots — coordinated hashtag campaigns directly preceded communal violence in multiple cities. Police had no real-time tool to detect coordinated inauthentic behavior or emerging offline threats from online content patterns.

Build a Bob-powered OSINT tool that ingests a batch of mock social media posts, detects coordinated inauthentic behavior signals, classifies threat type (incitement / targeted harassment / organized misinformation), maps to IPC/BNS provisions, and generates a time-stamped threat brief with recommended escalation steps for law enforcement.

07 Missing Person Investigation Assistant

Real Case: India has 80,000+ missing children annually (NCRB 2022). The first 24 hours are critical, but families waste time on paperwork while investigators lack a correlated view of social media signals, sighting tips, and movement patterns across data sources.

Build a Bob-powered case coordination tool that takes family-provided data plus mock investigator tip logs and CCTV sighting descriptions. Bob correlates inputs, generates a prioritized list of investigative leads with recommended next actions, drafts a public appeal notice, and auto-fills a police missing person case file.

08

Human Trafficking Risk Detection & Rescue Coordinator

Real Case: Muzaffarpur shelter home case (Bihar, 2018) — trafficking victims remained unidentified inside institutions for years. India is a major source, transit, and destination country (UNODC). Frontline workers have no structured identification tool.

Build a Bob-powered interview-based risk detection tool for NGO field workers and police. Bob conducts a structured 20-indicator interview, scores risk level, explains the flags to the interviewer, auto-generates an FIR-ready victim statement, and surfaces rescue organization contacts.

09

Child Online Safety Monitor & Reporting Guide

Real Case: POCSO cyber cases rose 400% post-COVID in India. NCMEC received 32M+ CSAM reports globally in 2022. Indian parents, teachers, and NGO workers have no accessible tool to identify grooming patterns early or file correct reports to NCRB's CyberTipline.

Build a Bob-powered early-warning and reporting tool for parents, teachers, and child protection workers. Bob identifies grooming warning signs from described behavioral changes or chat excerpts (mock), classifies risk level, guides evidence preservation without re-exposure, maps the case to POCSO / IT Act sections, and auto-completes a CyberTipline report ready for submission.

TRACK 4 · AI & PREDICTIVE

Intelligence, Prediction & Pattern Detection

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FIR Intelligence & Crime Pattern Detector

Real Case: UP Police's CCTNS system holds 3+ crore digitized FIRs with no NLP layer. Serial offenders like the Jamtara gang evaded detection for years because inter-district FIR connections were never surfaced. Pattern analysis is entirely manual.

Build a Bob-powered NLP intelligence tool that ingests a batch of mock FIR text samples. Bob categorizes each by crime type, extracts named entities (accused, location, MO, victim profile), detects repeat-offender signatures across FIRs, and generates a station-level crime trend summary with a flagged repeat-offender list.

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Drug Abuse Hotspot Prediction & Intelligence Report

Real Case: Punjab drug epidemic kills 15+ people daily. The 2022 SIT report identified 700+ peddler networks but had no predictive capability — enforcement was entirely reactive with no early-warning layer for emerging hotspots.

Build a Bob-powered narcotics intelligence tool for district officers. Bob takes structured district-level input data (drug seizure records, hospital OD admissions, arrest logs, demographic indicators — mock), performs multi-factor trend analysis, ranks districts by escalating-risk score, and generates a monthly narcotics intelligence report with recommended deployment actions.

Real Case: Delhi Police's ZIPNET has years of location-tagged crime data but no predictive layer. Beat officers patrol fixed routes regardless of emerging patterns. Bengaluru's 2023 predictive policing pilot reduced property crime by 18% in test zones.

Build a Bob-powered crime intelligence assistant for a district police station. Bob takes 6 months of mock crime data, identifies spatial-temporal concentration patterns, predicts the top 5 at-risk micro-zones for the coming week with probability rationale, flags seasonal or event-linked spikes, and generates a patrol redeployment recommendation brief for the Station House Officer (SHO).